Kasauti (कसौटी)

An Automated Regression Test Harness & Benchmark for Indic Voice AI

Gnani.ai Internship Challenge
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THE CHALLENGE: THE TELECOM REALITY

- **Acoustic Degradation:** Indian voice agents operate over 8 kHz band-limited, lossy telephony networks with heavy traffic, babble, and carrier compression.
- **Standard Metrics Fail:** Word Error Rate (WER) treats valid orthographic variations (e.g., *"₹5 लाख"* vs *"5,00,000 रुपये"*) as outright errors.
- **Manual QA Bottleneck:** Recording, crowdsourcing, and annotating test calls takes weeks, costs thousands, and cannot easily re-test new model checkpoints.

THE SOLUTION: CLOSED-LOOP KASAUTI

- **Synthetic Ground Truth:** Synthesizes seed sentences containing known phone numbers, PINs, rupee amounts, and timestamps using Gnani's **Timbre TTS**.
- **PSTN Channel Emulation:** Degrades audio through 300–3,400 Hz filtering, 8 kHz downsampling, G.711 μ-law quantization, and calibrated noise (20 dB down to 0 dB).
- **Semantic Value Scorer:** Normalizes Indic numeric ASTs to judge exact business meaning rather than superficial spelling.



EXPERIMENTAL RESULTS (156 CLIPS × 6 CONDITIONS)

Condition	Prisma v2.5	Prisma v3.0	Finding
Clean Audio	155 / 156	153 / 156	Identical failure traced to TTS defect
Phone Line (8 kHz μ-law)	155 / 156	154 / 156	Bandwidth compression alone has 0% loss
Phone + 10 dB Noise	151 / 156	153 / 156	Parity performance (p = 0.625)
Phone + 0 dB (Severe)	144 / 156	125 / 156	**v2.5 outperforms v3.0 (p < 0.001)**

KEY ENGINEERING DISCOVERIES

- **The 0 dB Noise Cliff:** Prisma v3.0 experiences heavy digit substitutions under intense noise (turning 1 into 8), whereas v2.5 preserves sequence length.
- **Attribution Isolation:** Identified a clean-audio bug in Timbre TTS where digit 8 was repeated 3 times instead of 4, proving STT was blameless.
- **Fragility Breakdown:** 82% of all failures occur in digit strings (phones/PINs); currency amounts and times are reinforced by surrounding context.
- **Hallucination Detection:** Caught low-SNR acoustic stress triggering models to hallucinate phantom tokens (predominantly *"था"*).

PRODUCTION FEASIBILITY & DEVELOPER VELOCITY

100%	₹0.00	146 / 146	p < 0.001
SYNTHETIC GROUND TRUTH	DETERMINISTIC CACHE RE-RUNS	OFFLINE UNIT TESTS (1.1S)	PAIRED EXACT SIGN TEST

GitHub Repository: **github.com/Tanmaysonkar/gnani-ai**

Live HTML Audit Report: `results/report.html` • Demo Video: 2m 32s